



United States — Monthly Intelligence Report

v4.2

Edition 005 — Deep-Dive Theme

Where the energy transition is outrunning the grid. edition · July 2026 · SSI v4.2 (refreshed June 2026)

CONTENTS

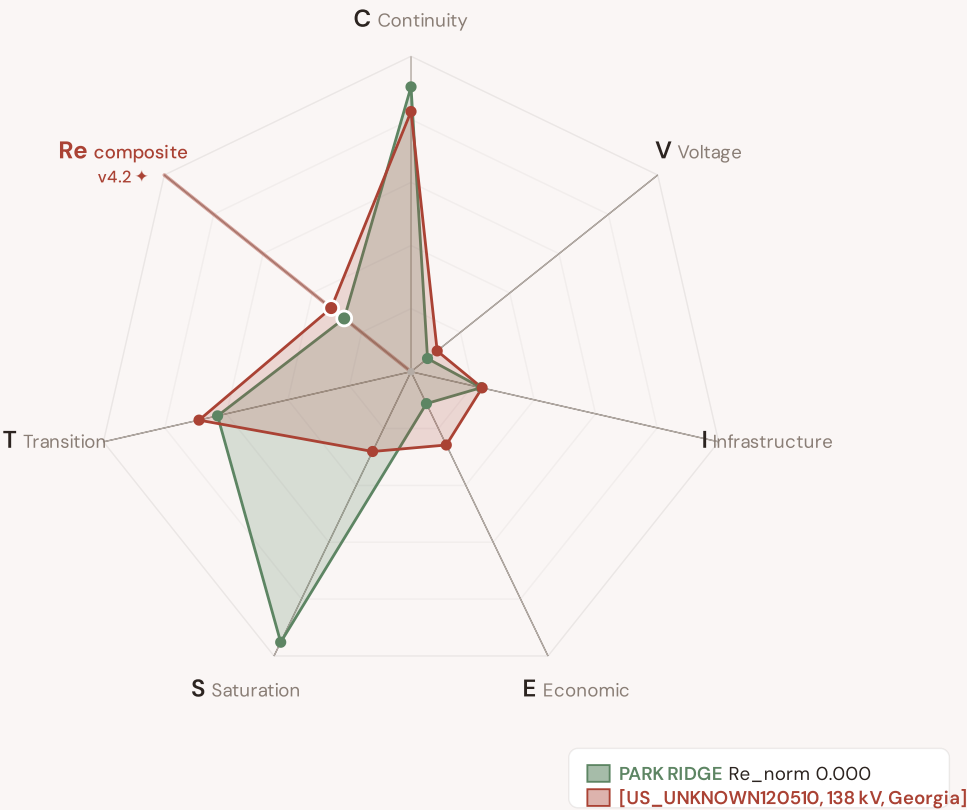
A Executive Summary	E European Context Card
B Cyber & Economic Exposure Monitor	F SSI Enhanced Neural Network
C Data Refresh & Changelog	F.5 Anti-maladaptation Gate (Radar 2 · W1–W10) v4.2
D Deep-Dive: Monthly Theme	G Looking Ahead

SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar (Re_norm ∈ [0, 1] via clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)). Two worked examples overlaid: **PARK RIDGE** (low-exposure reference, Re_norm 0.000) · **[US_UNKNOWN120510, 138 kV, Georgia]** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$, bounds [0.920, 1.787]. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared intelligence-sections.js and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

SUBSTATIONS SCORED

45,003

5,122 EHV · 22,484 HV ·
17,397 distribution-tier

GRID LINES MAPPED

68,400

~632,642 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

34

101 variables · zero
proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 45,003 substations and 68,400 grid lines spanning ~632,642 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering United States's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

TAP162292

Minnesota, Minnesota · 0 kV

0.387

Medium

0.133

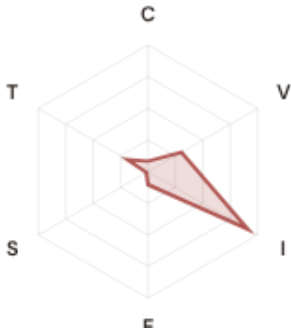
23th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



Re composite v4.2 7th axis

Re_raw 1.1694 · Re_norm 0.228

LOW EXPOSURE

ⓘ

COMPONENTS	MODIFIERS
C Continuity: 0.083 / 0.30 (28%)	R3 Consequence: 0.941 ↓ dampens
I Infrastructure: 0.911 / 0.25 (365%) ⚠	R4 Graph Criticality: 0.992 → neutral
S Saturation: 0.015 / 0.20 (8%)	R6a Restoration: 1.033 ↑ amplifies
V Voltage: 0.305 / 0.10 (305%)	R6b Seismic: 1.013 ↑ amplifies
E Economic: 0.102 / 0.10 (102%)	R7 Digital Readiness: 1.003 → neutral
T Transition: 0.188 / 0.05 (376%)	

This month's spotlight — automatically selected for July 2026 — is **TAP162292** in Minnesota, Minnesota. With an R_final of 0.387 (Medium band, 23th fleet percentile), its risk profile is dominated by the T (Transition) component at 376% of its weight ceiling, followed by I (Infrastructure) at 365%. Modifiers collectively shift R_base by 0.9%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

3756

AVG R7 MODIFIER

0.9950

Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

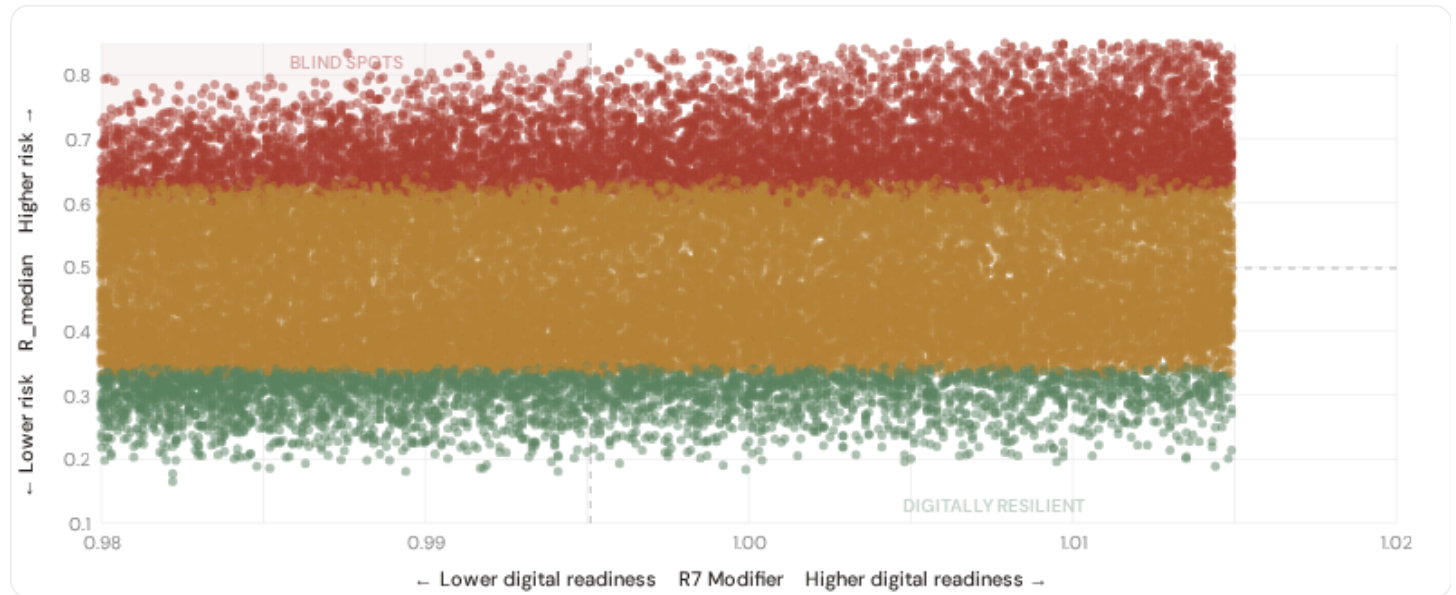
11,194

High/Critical + R7 < median

25.1% · R3 ≥ 0.97

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **3756 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 0.9951$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Florida (301)**, **Georgia (284)**, **Indiana (264)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

Capital-Intensive / High-Consequence

Est. VoLL: \$15–30/kWh (USD)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

11,194 substations (25.1%) High/Critical: **2261** (20.2%) Avg R: **0.503** Avg E: **0.245** / 0.10



Industrial / Medium-Large Enterprise

Est. VoLL: \$8–15/kWh (USD)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

11,240 substations (25.2%) High/Critical: **2567** (22.8%) Avg R: **0.509** Avg E: **0.250** / 0.10



Commercial / SME-Dense

Est. VoLL: \$3–8/kWh (USD)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

11,036 substations (24.7%) High/Critical: 2577 (23.4%) Avg R: 0.501 Avg E: 0.242 / 0.10



Light / Agricultural / Rural

Est. VoLL: \$1–3/kWh (USD)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

11,164 substations (25.0%) High/Critical: 3006 (26.9%) Avg R: 0.506 Avg E: 0.245 / 0.10

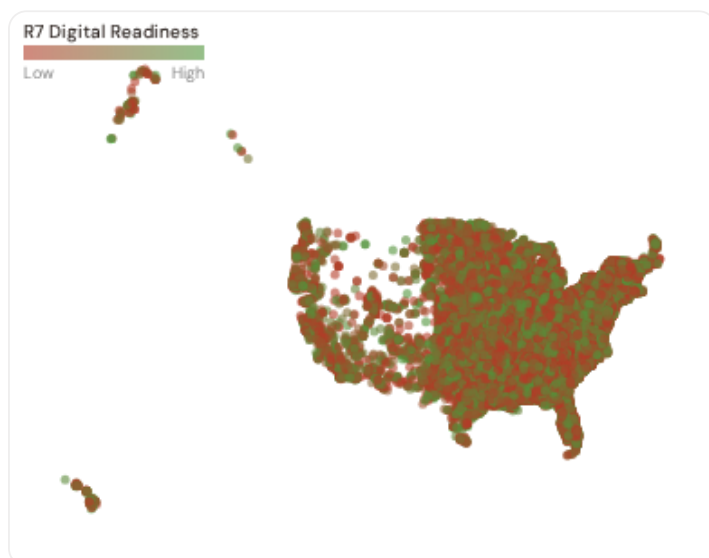


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **2261 substations** combine capital-intensive economic fabric ($R3 \geq 0.97$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **North Carolina (679), Alabama (675), Kentucky (537)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 11.7%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify provinces combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	New Hampshire	0.9922
2	Alaska	0.9935
3	Arizona	0.9937
4	Connecticut Δ high R	0.9937
5	Washington Δ high R	0.9939
6	Utah Δ high R	0.9944
7	Illinois Δ high R	0.9945
8	Hawaii Δ high R	0.9946
9	Mississippi	0.9946
10	Arkansas	0.9946
11	Pennsylvania Δ high R	0.9947
12	Indiana Δ high R	0.9947
13	Florida Δ high R	0.9948
14	Delaware Δ high R	0.9948
15	South Dakota	0.9948

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **New Hampshire** has the lowest average R7 (0.9922), while **District of Columbia** leads at 1.0012 — a spread of 0.0090 across the R7 range. **14 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

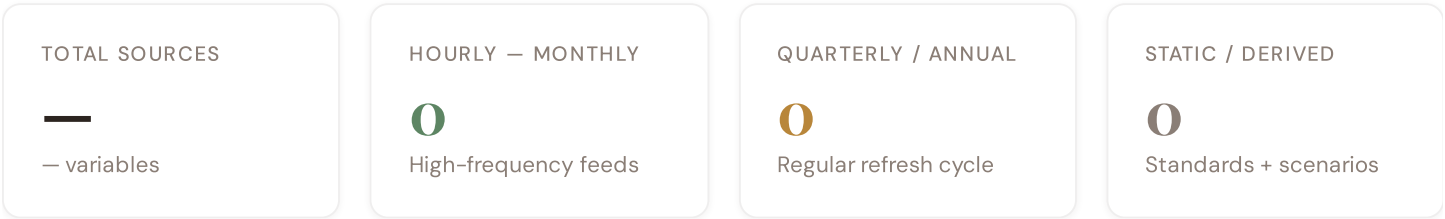
B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (July 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9950, median = 0.9951; 0 substations classified HIGH cyber-exposure; 3756 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.



Data Source Registry — July 2026

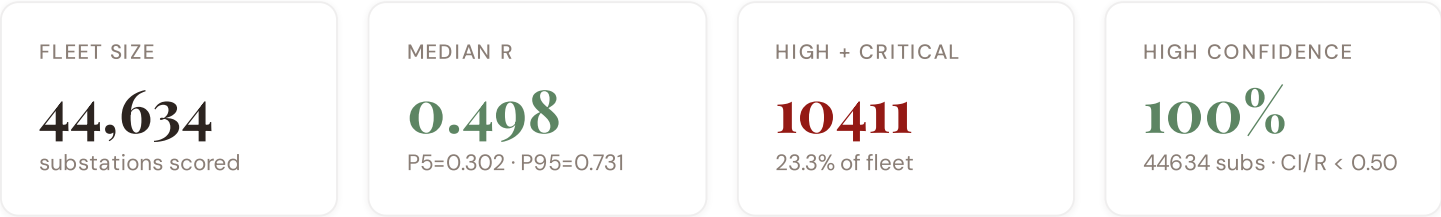
OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
BLS-LM	BLS Labour Market Statistics	MONTHLY	County	1 var
DOE	DOE/EIA (Department of Energy)	QUARTERLY	County	5 vars
EIA	EIA (Energy Information Administration)	ANNUAL	County	8 vars
Census	Census Bureau (US Census Bureau)	ANNUAL	County	8 vars
NERC	NERC (North American Electric Reliability Corp)	ANNUAL	Regional	2 vars
FEMA	FEMA / EPA (Federal Emergency Management Agency)	ANNUAL	County	4 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
BLS	BLS (Bureau of Labor Statistics)	ANNUAL	County	3 vars
NTIA	NTIA Broadband Map	ANNUAL	County	2 vars
CISA	CISA (Cybersecurity and Infrastructure Security Agency)	ANNUAL	National	2 vars
FEMA-FR	FEMA Flood Risk / Hazard Mitigation	ANNUAL	County	2 vars
NRC	NRC (Nuclear Regulatory Commission)	ANNUAL	Site-level	1 var
BEA	BEA (Bureau of Economic Analysis)	ANNUAL	County	3 vars
EIA-SS	EIA Statistical Series	ANNUAL	State/County	2 vars
FRA	FRA (Federal Railroad Administration)	ANNUAL	County	1 var
DOT	DOT/FHWA (Dept of Transportation)	ANNUAL	County	1 var
FERC	FERC Monitoring Report	ANNUAL	Regional	2 vars
PUC	State PUCs (Public Utility Commissions)	ANNUAL	Service Territory	2 vars
CESER	DOE CESER (Cybersecurity, Energy Security, Emergency Response)	ANNUAL	Regional	2 vars
Eurostat	Eurostat Energy Statistics	ANNUAL	EU Regional	3 vars
JRC	JRC European Commission	ANNUAL	European	2 vars
USGS	USGS (US Geological Survey)	STATIC	County	3 vars
CDS	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars

DIM	Dimovski et al. (2025)	STATIC	County	3 vars
IEEE	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
USGS-SH	USGS Seismic Hazard Assessment	STATIC	County	1 var
Census-GEO	Census Bureau Geography Registry	STATIC	County	1 var
USGS-HZ	USGS Hazard Assessments	STATIC	~5 km	2 vars
ISO9223	ISO 9223 Corrosion	DERIVED	County	1 var
EIA-ISO	EIA/ISO-RTE (PJM/CAISO/MISO) — Data Portal	HOURLY	Balancing Authority	3 vars
OM	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
NOAA	NOAA/NWS (National Oceanic and Atmospheric Admin)	HOURLY	~1 km	2 vars
NOAA-NWS	NOAA/NWS (National Weather Service)	HOURLY	Station	3 vars
ENTSO-E	ENTSO-E Transparency	HOURLY	Cross-border	2 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot



BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **44,634 substation scores remain unchanged** this edition. The fleet median R of 0.498 (mean 0.505) reflects a distribution skewed toward the lower bands, with 10.4% in Low and 66.2% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourgouin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across P0, P1, and P2 releases

V42-1	NEW	R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention)	\$5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])	\$5
V42-3	NEW	R6e_winter — Bourgouin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])	\$5
V42-4	NEW	R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])	\$5
V42-5	NEW	R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})	\$5
V42-6	NEW	R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])	\$5
V42-7	NEW	Re_composite — bounded resilience scalar $Re_{raw} = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$; $Re_{norm} = clip((Re_{raw} - 0.920)/(1.787 - 0.920), 0, 1)$	\$7
V42-8	NEW	R7_SFDR_PAI_Infrastructure_axis — added to ESG Radar (Re_{norm} proxy under Convention #7 Data-Layer Anchoring)	\$14
V42-9	NEW	W1-W10_audit-state_mesh — 5 Published / 1 Published-baseline▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6)	\$4
V42-10	CHANGE	95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)	data
V42-11	CHANGE	Substation fleet: 45,003 substations (per OSM canonical extract + national TSO/DSO registry cross-validation)	data

SECTION D — MONTHLY DEEP-DIVE

Monthly Deep-Dive Theme

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North-South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress-Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Theme Context

THEME-RELEVANT
SUBSTATIONS

353

47 EHV · 306 HV

HIGH BAND

236

66.9% of corridor

CORRIDOR MEDIAN R

0.667

Fleet median: 0.498

WORST PROVINCE

Washington

Avg R: 0.633 · 353 substations

The Washington hosts **353 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **236 substations (66.9%)** are classified High, with only **10** in the Low band. 83% of corridor substations score above the national fleet median of 0.498. The corridor median R of 0.667 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
PICNIC POINT	Washington	0.896	High	0.992	0.728	0.697	1.000	1.000	0.770	C, S	
TAP209806	Washington	0.869	High	1.000	0.429	0.734	1.000	1.000	0.929	C, S, T	
TUKWILA STATION	Washington	0.854	High	0.904	0.467	0.719	0.814	1.000	0.582	C, S	
KENMORE	Washington	0.851	High	0.980	0.832	0.781	1.000	1.000	0.863	C, I, S, T	
SNOHOMISH PUD	Washington	0.851	High	1.000	0.416	0.697	1.000	1.000	0.697	C, S	
PRESIDENT PARK	Washington	0.838	High	1.000	0.604	0.660	1.000	1.000	0.962	C, S, T	
OLYMPIC VAIL PIPELINE	Washington	0.833	High	0.834	0.803	0.578	1.000	0.932	0.883	C, S, T	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
MARTHA LAKE	Washington	0.826	High	0.914	0.553	0.775	1.000	1.000	0.728	C, I, S	
UNKNOWN202934	Washington	0.819	High	0.900	0.467	0.606	1.000	1.000	0.978	C, S, T	
TAP206097	Washington	0.812	High	0.918	0.637	0.777	1.000	0.986	0.790	C, I, S	
... 343 additional substations — explore full corridor in Digital Twin →											

D.3 Component Analysis

COMPONENT AVERAGES — WASHINGTON VS FLEET			MODIFIER AVERAGES — WASHINGTON VS FLEET		
C — Continuity	0.643 vs 0.490 (+31%)		R3 Consequence	0.955	fleet 0.955 ↓
I — Infrastructure	0.393 vs 0.539 (−27%)		R4 Graph Crit.	0.978	fleet 0.980 ↓
S — Saturation	0.807 vs 0.348 (+132%)		R6a Restoration	1.018	fleet 1.020 ↑
V — Voltage	0.496 vs 0.382 (+30%)		R6b Seismic	0.998	fleet 1.000 →
E — Economic	0.964 vs 0.246 (+292%)		R7 Digital Read.	0.994	fleet 0.995 →
T — Transition	0.645 vs 0.466 (+38%)				

The dominant risk driver across the Washington corridor is **T (Transition)**, averaging 0.645 against a fleet average of 0.466 — occupying 1290% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.964 vs fleet 0.246 (964% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.955 (fleet: 0.955), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 0.998, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Washington

353 subs · avg R = 0.633 · H:236 M:107

The region regions-level breakdown reveals significant intra-regional variation. **Washington** leads with a mean R of 0.633 across 353 substations, while **Washington** scores 0.633 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Washington is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

195 Washington substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Washington sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic

exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

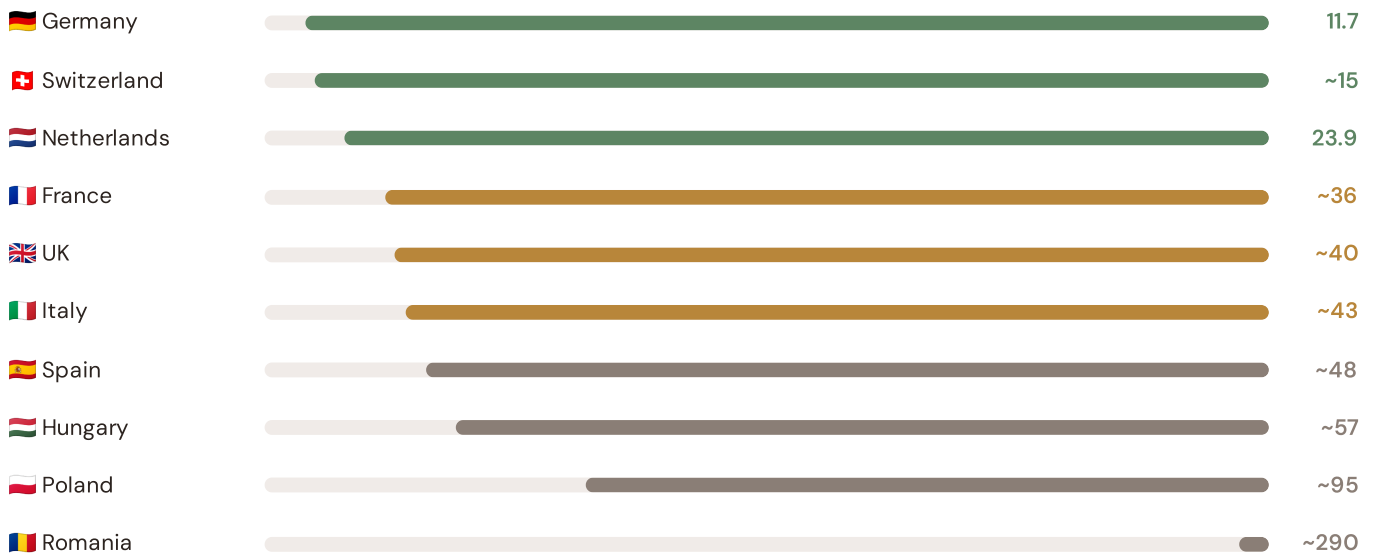
SECTION E — RECURRING MODULE

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on United States's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



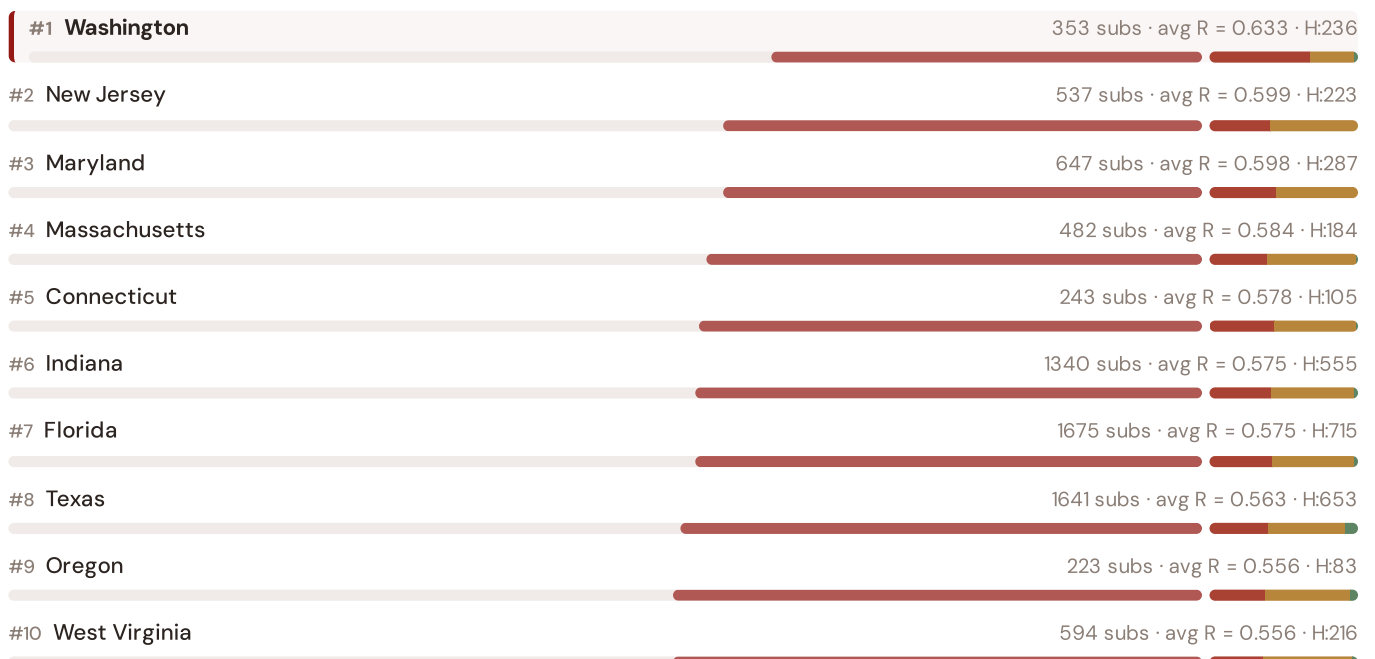
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

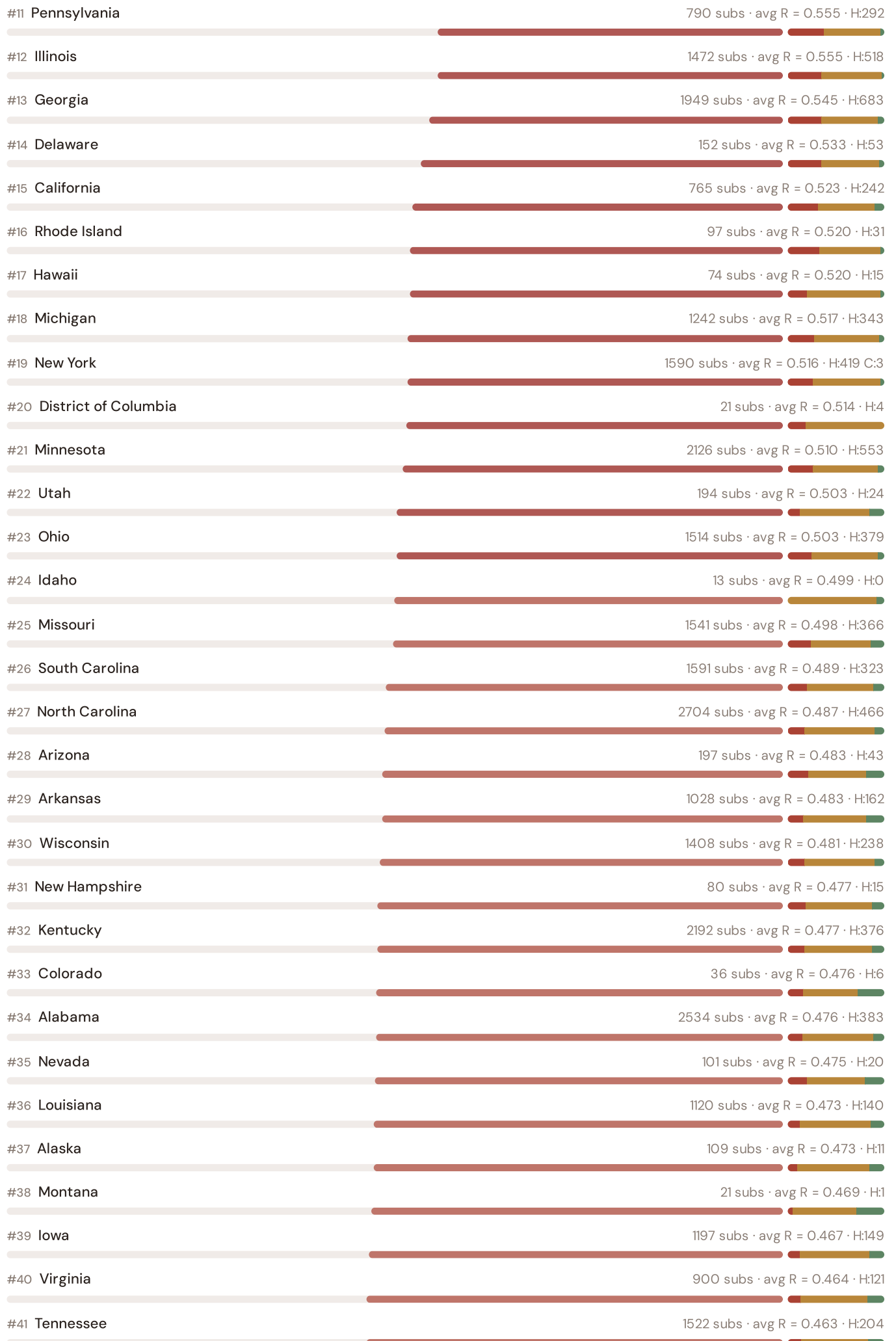
This month's key insight: The Washington corridor deep-dive shows **301 substations** (of 353) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.643 is **1.3× the fleet average** of 0.490. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

States Risk Ranking — All 51 American States

Where does this corridor sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.







The Washington ranks **#1 of 52 region regions** by mean R_{median} , placing it among the upper tier of national risk. The three most stressed are Washington, New Jersey, Maryland, while the three most resilient are Unassigned, South Dakota, Nebraska. 21 of 52 score above the fleet average of 0.505. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

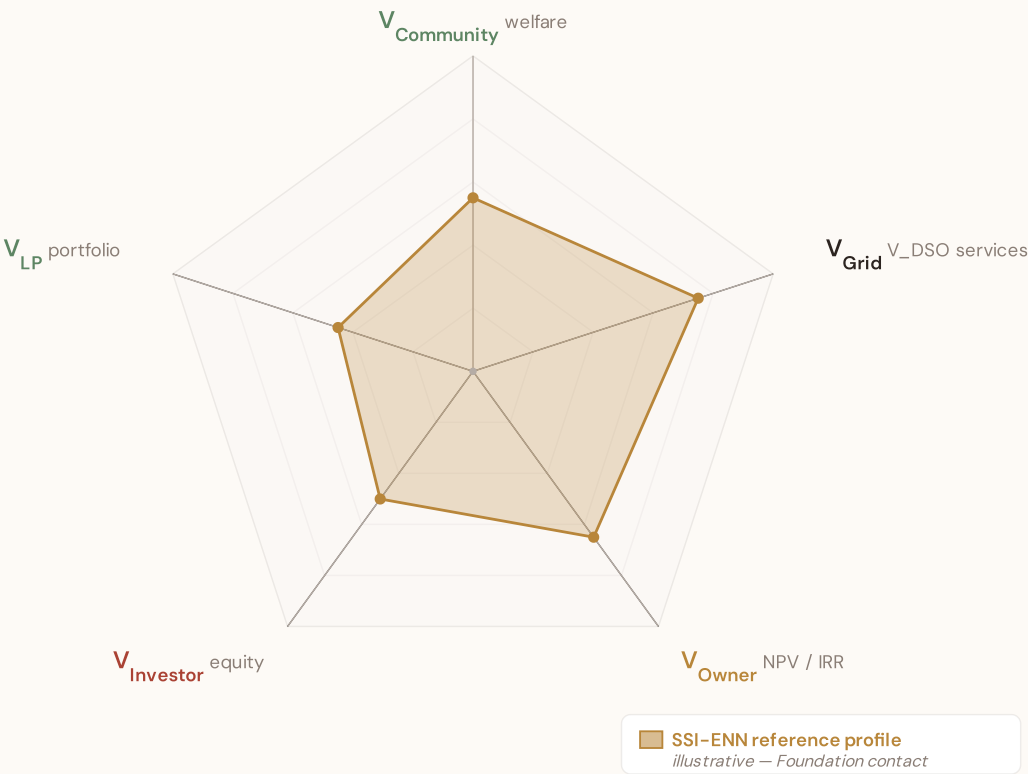
SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): **V_{Community}** (welfare avoided outages, energy-poverty relief, employment) · **V_{Grid}** (V_{DSO} — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · **V_{Owner}** (NPV / IRR / payback per asset class) · **V_{Investor}** (risk-adjusted equity return at the financing tier) · **V_{LP}** (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_{DSO} captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{Community}$ + V_{Owner} are co-secondary. $V_{Investor}$ + V_{LP} reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics

LAYER 3 §0.5.5 · July 2026

Uncertainty Quantification

Investment-grade predictions require calibrated confidence intervals, not just point estimates

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (pinball loss) and conformal prediction (distribution-free guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI_{width}, P5/P95) serve as the ground truth for calibration. This calibration is verified per country, per voltage class, and per risk band.

KEY CONCEPTS

- Quantile regression (pinball loss) + conformal prediction
- Calibrated against SSI Monte Carlo ground truth
- Per-country, per-voltage, per-band calibration verification
- Investment-grade: P10/P50/P90 for all outputs

LIVE SSI DATA CONNECTION

The training set comprises 44,634 substations with complete feature vectors. Average CI_width of 0.147 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 4658 Low, 29565 Medium, 10408 High, 3 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 2** Total Cost of Ownership Model — *Stochastic cost modelling — because BESS economics are not deterministic*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework. The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill
W1 · W2 · W3 · W4 · W8
Peer-reviewed methodology operational at fleet scale.

Published-baseline (1) ▲ — solid fill + chevron corner marker
W9 — Supply Security
Peer-reviewed academic basis (Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).

Baseline-only (1) — striped fill
W5 — Digital Inclusion
Foundation methodology under development; baseline value defensible at v4.2.

Commercial (3) — dashed-faded
W6 · W7 · W10
Commercial deliverables — please contact the Foundation.

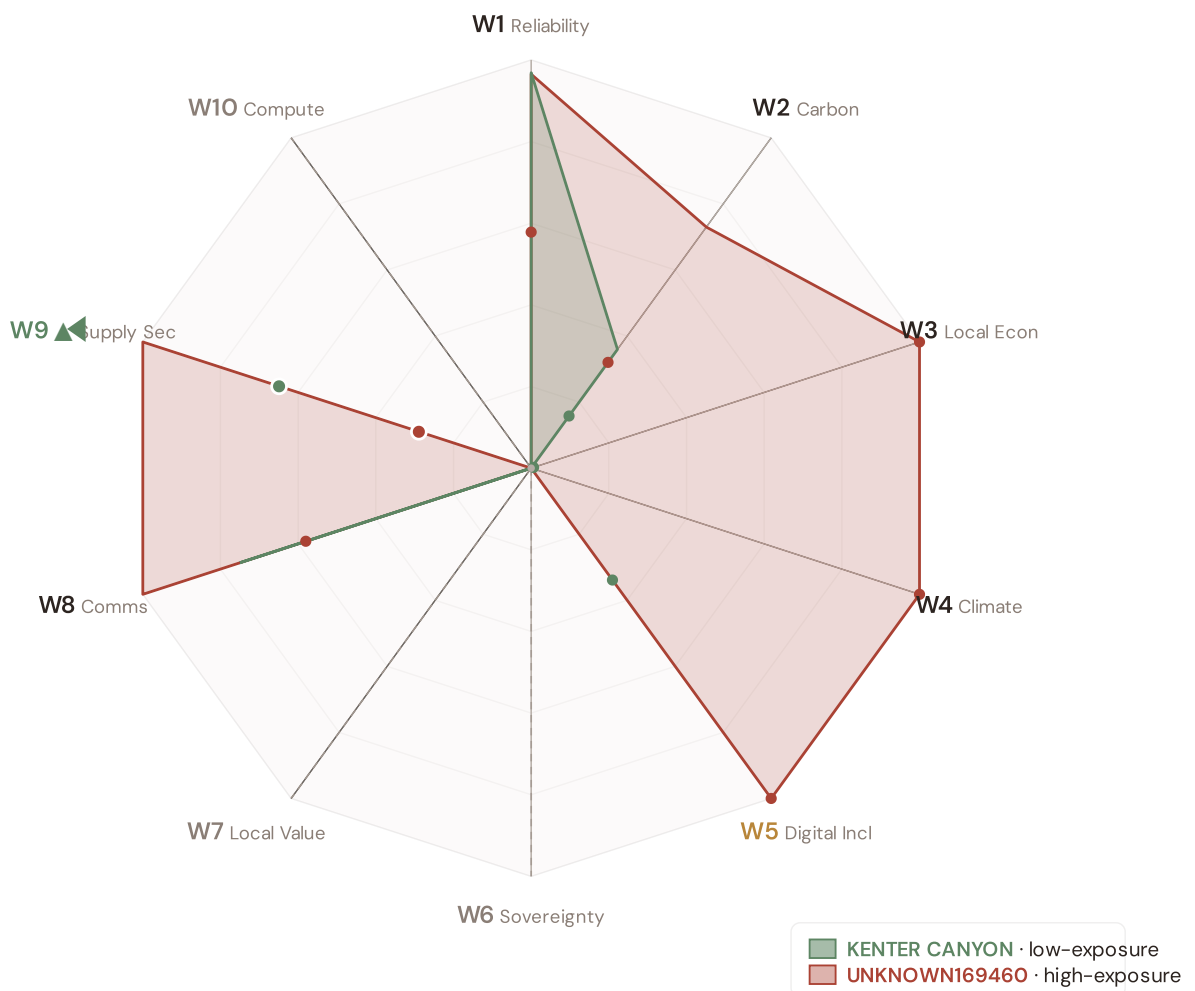
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

v4.2 methodology disclosure — per-country normalisation

Across the United States, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to **[0, 1]** via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (SSI_v4_0 Italy/.../scoring-it/normalise.py); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10–d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **PARK RIDGE** (low-exposure reference, Re_norm 0.000) and **[US_UNKNOWN120510, 138 kV, Georgia]** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



KENTER CANYON polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. UNKNOWN169460 polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.

W1–W10 axis enumeration with audit-state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	Published	$C \times E2_{\text{local}} \times \text{population} (SAIDI++-POV)$
W2	Carbon	Published	$T1 \times \text{zonal MEF} \times (V \cdot \text{deg} / 100) (+EEA)$
W3	Local Economic	Published	$\text{norm}(R10_{\text{just}}) - \text{EU-SILC} + \text{OIPE energy-poverty}$
W4	Climate Resilience	Published	$(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - \text{v4.2 resilience modifier family}$
W5	Digital Inclusion	Baseline-only	$\text{norm}(R8_{\text{adapt}}) - \text{bulk SM}++\text{DESI}$
W6	Sovereignty	Commercial	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	Commercial	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	Published	$C \times p_{\text{crit_10yr}} \times \text{population (Markov 10-yr critical probability)}$
W9 ▲	Supply Security	Published-baseline ▲	$0.60 \times BC_{\text{pct}} + 0.30 \times \text{bridge} + 0.10 \times (1 - \text{deg} / 20) - \text{OSM power-graph betweenness centrality} + \text{bridge identification} + \text{degree-based redundancy}$
W10	Compute Multiplier	Commercial	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published-baseline — mandatory transparency tooltip

W9 — Supply Security (Published-baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1-W10 per-substation baseline values for two anchor cases — KENTER CANYON (low-exposure reference) and UNKNOWN169460 (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max `Re_norm`) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 03

Capital Metropolitan

Deep-dive into Region O2's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Capital-intensive tier + flood overlay.

NEXT DATA REFRESH

Q4 2026 — o Quarterly Sources

none are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **EIA (Energy Information Administration)** (8 variables → C1–C4 (SAIDI/SAIFI/CAIDI), quality regulation).

FLEET SPOTLIGHT

o Bridge Substations — Single Points of Failure

O substations are topological bridges — their failure disconnects part of the network. Of these, **O** are classified High or Critical. These are the fleet's single points of failure: high graph centrality (R4) combined with elevated risk makes them priority candidates for both grid reinforcement and BESS deployment.

Edition 03 will be published on the second Thursday of August 2026. Deep-dive region: **Region 02**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an American utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 005

Published 9 July 2026 · SSI Index **v4.2** (refreshed June 2026) · 45,003 substations · 68,400 grid lines · 6 components + **Re composite** (Radar 17th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5 · W1-W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.

